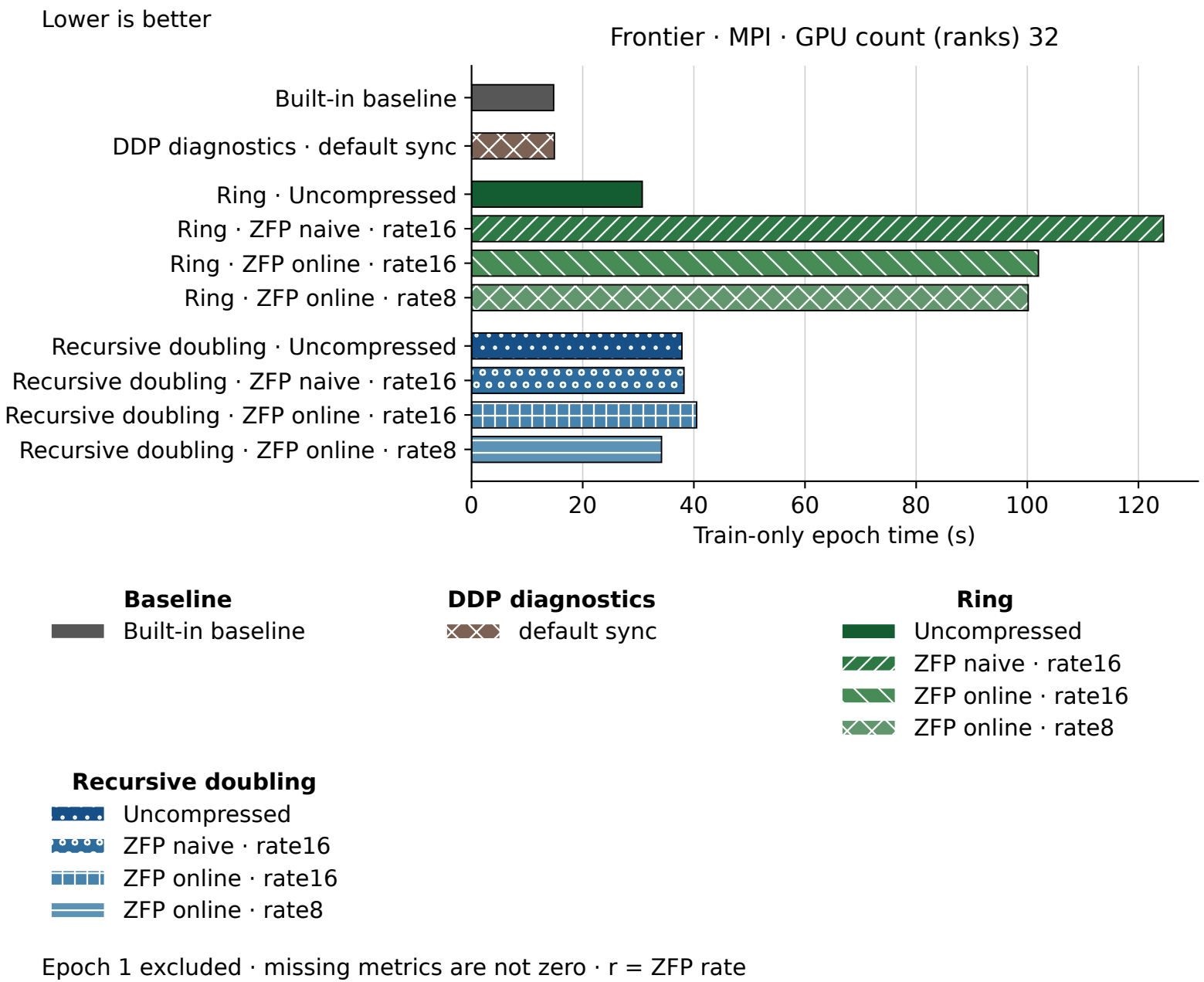

WideResNet 224×224 · strong scaling · effective global batch 128
Lower is better
Frontier · MPI · GPU count (ranks) 32
Built-in baseline
DDP diagnostics · default sync
Ring · Uncompressed
Ring · ZFP naive · rate16
Ring · ZFP online · rate16
Ring · ZFP online · rate8
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Recursive doubling · ZFP online · rate16
Recursive doubling · ZFP online · rate8
0
20
40
60
80
100
120
Train-only epoch time (s)
Baseline
Built-in baseline
DDP diagnostics
default sync
Ring
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Recursive doubling
Uncompressed
ZFP naive · rate16
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate